

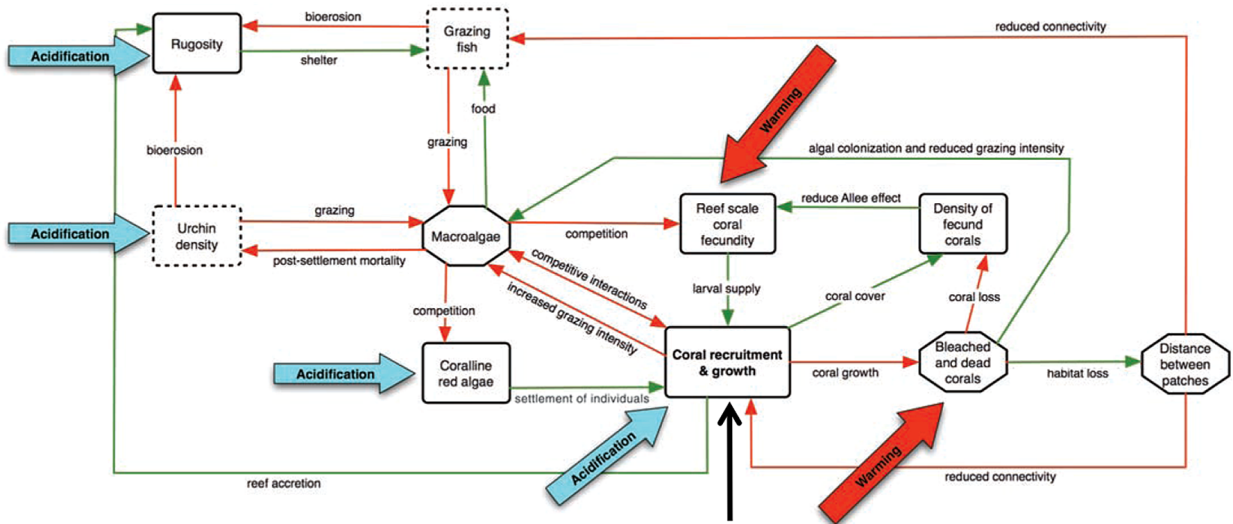
Read each question carefully and completely. Be sure your answer addresses the question being asked. Use complete sentences.

- (10 pts) 1. Write the equilibrium equation for the carbonate buffering system in marine ecosystems. Use this equation to explain how the increase in atmospheric CO_2 increases ocean acidity.

- (5 pts) 2. Write the equilibrium equation for CaCO_3 chemistry.

(10 pts) 3. Explain how the increased acidity (from the equilibrium equation in question 1) interacts with CO_3^{2-} (from the equilibrium equation in question 2) to reduce CaCO_3 availability to marine animals. Include the proper chemical equation as part of your answer

(5 pts) 4. What is the minimum value of Ω necessary for carbonate skeleton formation? What is the optimum value?



- (20 pts) 5. Using the figure above (also on screen), clearly and thoroughly explain how increased ocean acidity will alter connectivity and fecundity across a large coral reef. You must include coral recruitment and growth (indicated by the black arrow), distance between patches and coral fecundity. Be sure to include both positive and negative relation among these factors. Do not include unnecessary factors and relationships.

(5 pts) 6. Briefly explain the three factors that must be considered to identify a potential biodiversity hotspot for conservation efforts.

(10 pts) 7. Explain the concept and goals of maximum sustainable yield (MSY) as it relates to commercial fisheries. What population / life history parameters are required to estimate MSY? Why is MSY hard to implement in practice?

(5 pts) 8. MSY is based on a population logistic growth curve. Draw the logistic growth curve, and then draw an arrow to indicate the point along the growth curve that provides the theoretical maximum sustainable yield.

(10 pts) 9. Explain at least two reasons a manager must understand how genetic variation is distributed across the entire range of a species that is being considered for conservation management or reintroduction into a particular area.